

EE200: Signals, Systems and Networks

Course Project — Q2: The Midnight Episode

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The goal of the problem is to detect the point where ECG changes from normal rhythm to arrhythmic rhythm. For this we can use healthy heartbeat as a template and compare with others. Also we explore the frequency domain of ECG signal, spectrogram and sampling effects

(a) Reading the signal

(i) **Clip duration:**

$$\frac{N}{f_s} = \frac{5000}{250} = 20 \text{ s}$$

So, clip is 20 sec long.

(ii) A healthy beat repeats every $T = 0.8$ s. So,

$$\text{Heart Rate} = \frac{60}{0.8} = 75 \text{ bpm}$$

The number of samples in one beat is

$$f_s T = 250 \times 0.8 = 200 \text{ samples}$$

so patient's heart rate is 75 beats per min and 200 samples a healthy beat occupies.

(iii) **Fundamental frequency:** As the fundamental frequency is inverse of fundamental time period, which is 0.8s here, so

$$f_0 = \frac{1}{T} = \frac{1}{0.8} = 1.25 \text{ Hz}$$

hence f_0 is 1.25 Hz.

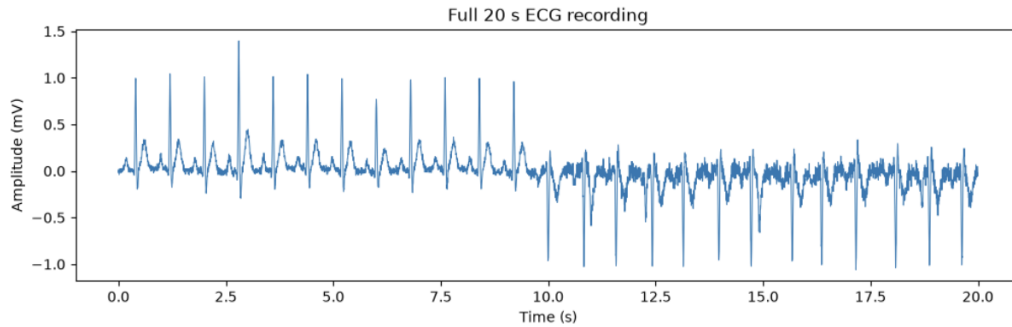


Figure 1: First ~9.6 s are regular, after that beats become irregular.

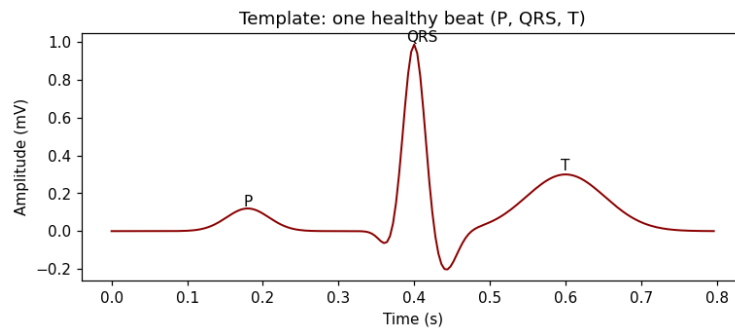


Figure 2: Healthy-beat template, with the P, QRS and T components labelled.

(b) Healthy heart in frequency domain

- (i) **Magnitude** The healthy ECG is nearly periodic, so its magnitude spectrum is not a smooth continuous curve. Instead, it consists of discrete spectral lines at the fundamental frequency f_0 and its harmonics i.e multiple of f_0 .

(ii) **Higher-frequency content**

The QRS is responsible for most of the higher-frequency content of the ECG. As it is very narrow and changes rapidly with time. Rapid changes in a signal require higher-frequency components, whereas the smoother P and T waves mainly contribute to lower-frequency content.

(iii) **Rise of heart rate**

If the heart rate increases to 150 bpm, then the time period becomes

$$T = \frac{60}{150} = 0.4 \text{ s}$$

The fundamental frequency is

$$f_0 = \frac{1}{T} = \frac{1}{0.4} = 2.5 \text{ Hz}$$

So the fundamental frequency doubles from 1.25 Hz to 2.5 Hz.

Spectrum occurs at multiple of f_0 so the space between the spectrum component also gets doubled.

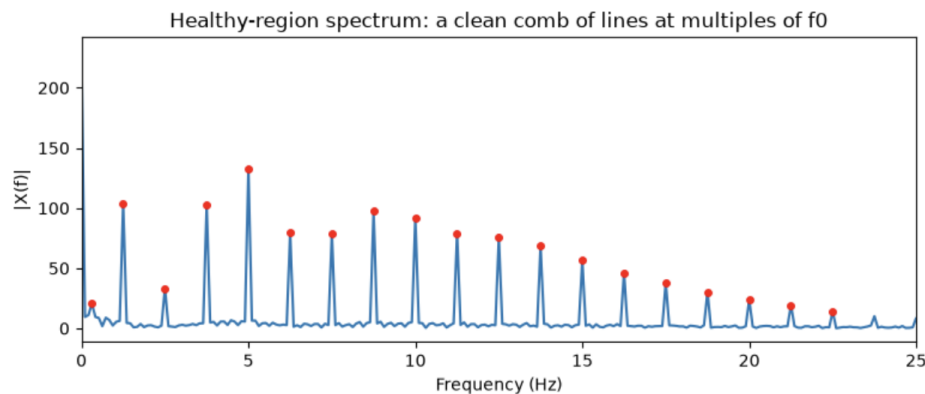


Figure 3: Spectrum of healthy periodic beats, at integer multiples of $f_0 = 1.25$ Hz.

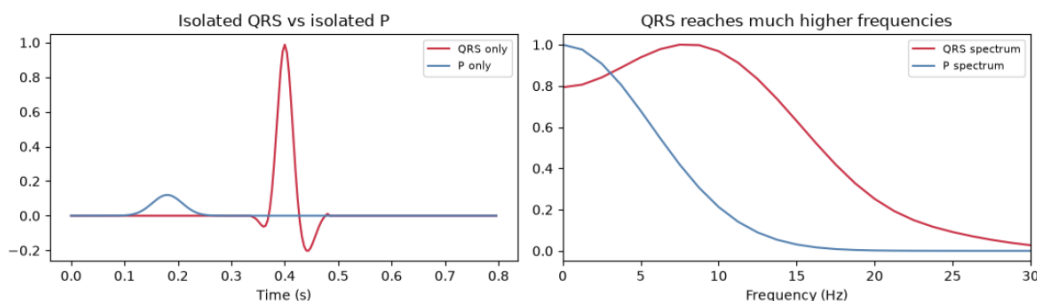


Figure 4: Isolated QRS spike vs. isolated P wave (left), and their normalized spectra (right)

(c) Cutting a heartbeat

(i) Window width and location

One healthy beat lasts for

$$f_s T = 250 \times 0.8 = 200 \text{ samples.}$$

So, the window should be 200 samples wide. She should place in the healthy part of the recording before the arrhythmia begins.

(ii) Effect of different window sizes

For 80-sample window, the whole heartbeat cannot fit inside the window, so important parts of the P–QRS–T waveform are missed. With a 600-sample window, it captures more than one heartbeat. Hence, the part of segment is not template of a single beat. So, a window of 200 samples better for template.

- (iii) **Time-frequency trade-off** Important part of heartbeat can be missed if we make the window as short as possible. The window should be long enough to capture the complete P–QRS–T waveform but not so long that it includes others beats next to it or before it.

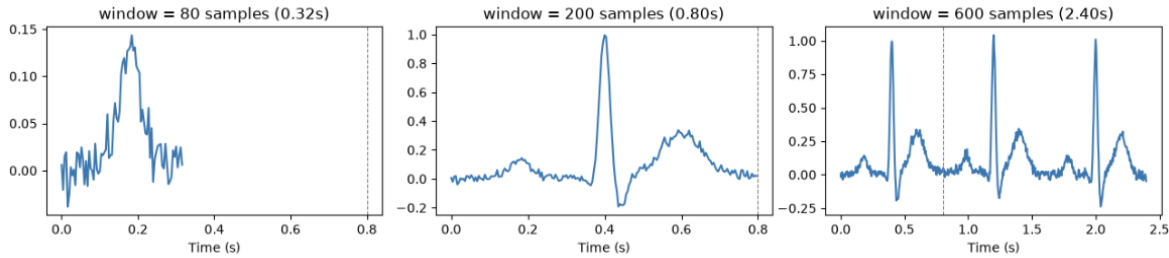


Figure 5: Comparison of three window lengths

(d) Match the template

$$\rho(m) = \frac{\sum_k t[k] x[m+k]}{\|t\| \|x_m\|}$$

(i) Range of correlation

$\rho(m)$ satisfies:

$$-1 \leq \rho(m) \leq 1.$$

Value of $\rho(m)$ when tends to 1 means nearly perfect match between template and the ECG. A value close to 0 means similar, while a value close to -1 means an inverted version of the template.

(ii) Normalization

ECG signals may vary in amplitude. So by doing normalization, we can remove the effect of signal amplitude and thus we can make it completely depend on waveform shape.

If a healthy beat is twice as tall as the template, i.e.

$$x[m+k] = 2t[k],$$

then the unnormalized score is:

$$\sum_k t[k] x[m+k] = 2 \sum_k t[k]^2,$$

which is twice the original value, even the beat shape is unchanged. After dividing it by the energies, the normalized correlation remains close to 1 for both signals.

(iii) Inverted beat

If an abnormal beat is an inverted version of the template, then

$$x[m+k] \approx -t[k].$$

So, the normalized correlation is expected to be close to

$$\rho(m) \approx -1.$$

As healthy beat has correlation close to 1 and this has -1 so it is easy to detect the inverted beats easily.

(e) Onset detection & the spectrogram

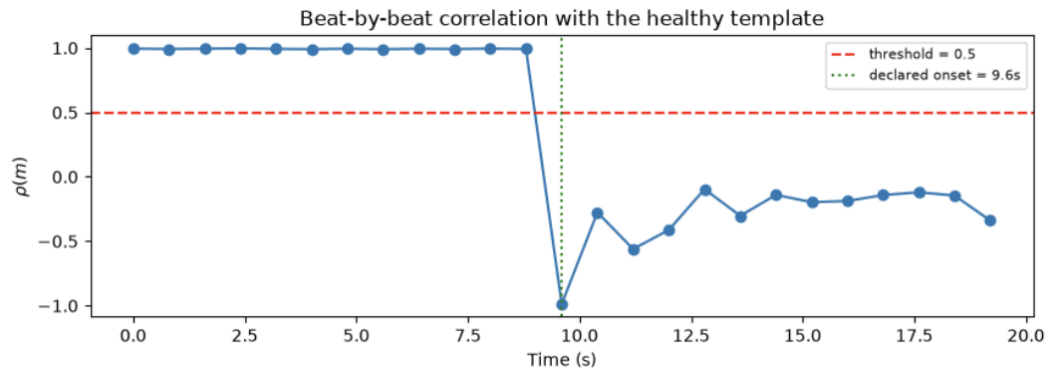


Figure 6: Beat-by-beat correlation with the template. Starting beats has $\rho = 1$ and then becomes nearly -1

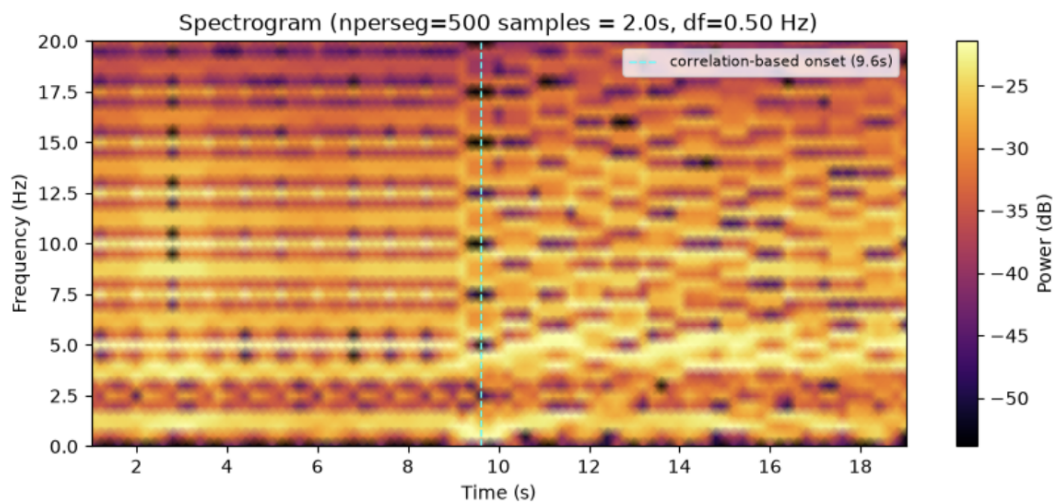


Figure 7: Horizontal harmonic bands before the onset; patchy, broken bands after.

- (i) **Onset** Maya can use that she should declare the onset when the correlation score $\rho(m)$ falls below a chosen threshold.

If the threshold is too high, then small normal variations between healthy beats may be detected as arrhythmia which can detect wrongly. If the threshold is too low, the detection may be delayed and thus many abnormal beats may be missed.

- (ii) **Spectrogram** In the healthy region, spectrogram is showing the horizontal line at fundamental freq and its harmonic freq which tells that there is a regular hearbeats.

In the arrhythmia region, these bands become weaker, distorted, or broken as the heartbeat is no longer periodic. Fig 7 shows this clearly.

- (iii) **Correlation vs spectrogram** The spectrogram uses a sliding window, so changes are spread across the window and the onset time may appear slightly shifted.

The correlation plot has better time resolution so it is more suitable for identifying the exact beat where arrhythmia first appears.

(f) Sampling & aliasing

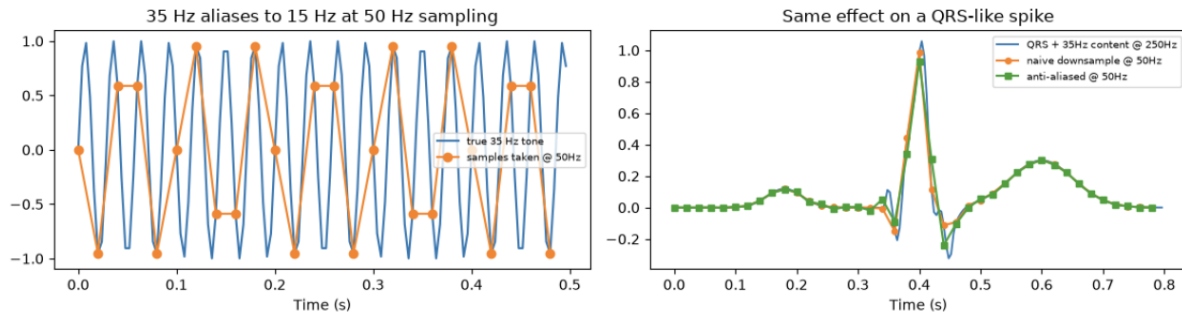


Figure 8: Left: 35 Hz tone Sampled at 50 Hz with no anti-aliasing filter. Right: the same folding effect applied near a QRS-like spike.

- (i) **Nyquist** Nyquist theorem says that the sampling frequency must be at least twice the highest frequency present in the signal. since the ECG contains freq upto 40Hz, so :

$$f_s \geq 2 \times 40 = 80 \text{ Hz.}$$

So minimum 80Hz is required to avoid aliasing.

- (ii) **Effect of aliasing** At 50 Hz, the sharp QRS spikes will not be captured accurately and their shape may become distorted.

As Maya's detector depends on matching the shape of each beat with a healthy template, this distortion can lead to incorrect detection of arrhythmia.

- (iii) **Reducing aliasing** The fix is to use a low-pass filter before sampling so that high-frequency components are removed before the signal is recorded.

The cost is that some important details of the ECG are lost. The QRS spikes become less sharp and the recorded signal is not exactly the same as the original ECG.

(g) Prototyping the detector in code

(i)

```
def find_onset(ecg_signal, template, threshold):
    L = len(template)
    n_beats = len(ecg_signal)
    t_norm = np.linalg.norm(template)

    for i in range(n_beats):
        m = i * L
```

```

    seg = ecg_signal[m:m + L]
    denom = t_norm * np.linalg.norm(seg)
    score = np.dot(template, seg) / denom if denom != 0 else 0.0
    if score < threshold:
        return m
return -1

```

(ii) At threshold of 0.5 the function will give:

$$m = 2400.$$

As the sampling frequency is 250 Hz, so:

$$t = \frac{2400}{250} = 9.6 \text{ s.}$$

So, the detector identifies the onset of arrhythmia at 9.6 s.

(h) Visualizing the spectrogram

```

import numpy as np
import matplotlib.pyplot as plt
from scipy import signal

nperseg = 500
f, t_spec, Sxx = signal.spectrogram(ecg_signal, fs=fs, nperseg=nperseg,
                                     noverlap=nperseg - 50)
plt.pcolormesh(t_spec, f, 10*np.log10(Sxx + 1e-12), shading='gouraud', cmap='inferno')
plt.ylim(0, 20)
plt.xlabel("Time (s)"); plt.ylabel("Frequency (Hz)")
plt.title(f"ECG spectrogram (nperseg={nperseg})")
plt.colorbar(label="Power (dB)")
plt.show()

```

Let us choose `nperseg=500` samples (2 s) which means 250 samples recorded per second. The corresponding frequency resolution is

$$\Delta f = \frac{f_s}{\text{nperseg}} = \frac{250}{500} = 0.5 \text{ Hz.}$$

This window size gives a clear spectrogram of the healthy ECG and also helps us observe the change when arrhythmia starts.